

Mark schemes

Q1.

(a) $y = e^x \rightarrow e^{2x} - 1$
Stretch (I)

scale factor $\frac{1}{2}$ (II)
 $I + (II \text{ or } III)$

M1

in x -direction (III)
 $I + II + III$

A1

Translation
Allow "translate"

E1

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

OE "1 unit down" etc

B1

4

(b) $x = 0 \quad y = 6 \quad \text{or} \quad (0, 6)$

Both coordinates must be stated, not simply 6 marked on diagram

B1

1

(c) (i) $e^{2x} - 1 = 4e^{-2x} + 2$
 $e^{4x} - e^{2x} = 4 + 2e^{2x}$
or $(e^{2x})^2 - e^{2x} = 4 + 2e^{2x}$
Multiplying both sides by e^{2x}

M1

$(e^{2x})^2 - 3e^{2x} - 4 = 0$
AG With no errors seen

A1

2

(ii) $(e^{2x} - 4)(e^{2x} + 1)$

$$(e^{2x} \pm 4)(e^{2x} \pm 1)$$

M1

$$x = \ln 2 \text{ or } \frac{1}{2} \ln 4$$

A1

Reject $e^{2x} = -1$ OE

Eg $e^{2x} > 0$, $e^{2x} \neq -1$, impossible etc

A1

3

(d) $\int (4e^{-2x} + 2) dx$ (I)

$$= \left[\frac{4e^{-2x}}{-2} + 2x \right]_0^{\ln 2}$$

I or II attempted and e^{-2x} or e^{2x} integrated correctly

M1

$$= \left(\frac{4e^{-2\ln 2}}{-2} + 2\ln 2 \right) - \left(\frac{4}{-2} + 0 \right)$$

F ['their $\ln 2$ ' from (c)(ii)] – F[0]

m1

$$= -\frac{1}{2} + 2\ln 2 + 2 = \frac{3}{2} + 2\ln 2$$

$$\int (e^{2x} - 1) dx$$
 (II)

$$= \left[\frac{e^{2x}}{2} - x \right]_0^{\ln 2}$$

Both I and II correctly integrated

A1

$$= \left(\frac{e^{2\ln 2}}{2} - \ln 2 \right) - \left(\frac{1}{2} - 0 \right)$$

$$= 2 - \ln 2 - \frac{1}{2} = \frac{3}{2} - \ln 2$$

$$A = \left(\frac{3}{2} + 2\ln 2 \right) - \left(\frac{3}{2} - \ln 2 \right)$$

Attempt to find difference of 'their I – their II'

B1ft

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

A1

Alternative

$$A = \int (4e^{-2x} + 2) dx - \int (e^{2x} - 1) dx$$

Condone functions reversed

(B1)

$$= \int_{(0)}^{(\ln 2)} (4e^{-2x} - e^{2x} + 3) dx$$

$$= \left[\frac{4e^{-2x}}{-2} - \frac{e^{2x}}{2} + 3x \right]_0^{\ln 2}$$

e^{2x} or e^{-2x} correctly integrated

(M1)(A1)

$$= \left(-2e^{-2 \ln 2} - \frac{1}{2}e^{2 \ln 2} + 3 \ln 2 \right) - \left(-2 - \frac{1}{2} \right)$$

Correct substitution of their $\ln 2$ from (c)(ii) into their integrated expression

(m1)

$$= 3 \ln 2 \text{ or } \ln 8 \text{ or } \frac{3}{2} \ln 4 \quad \text{OE}$$

CSO must be exact

(A1)

5

[15]

Q2.

(a) (i) $7x - 3 = A(3x - 2) + B(x + 1)$

M1

$$x = -1 \quad x = \frac{2}{3}$$

Substitute two values of x and solve for A and B

m1



$$A = 2 \quad B = 1$$

$$\text{Or solve } \left. \begin{array}{l} 7 = 3A + B \\ -3 = -2A + B \end{array} \right\} \text{ condone one error}$$

A1

Alternative:

By cover up rule

$$x = -1 \quad A = \frac{-7 - 3}{-5}$$

$$x = \frac{2}{3} \quad B = \frac{\frac{14}{3} - 3}{\frac{5}{3}}$$

$$x = -1 \text{ and } x = \frac{2}{3}$$

and attempt to find A and B

(M1)

$$A = 2 \quad B = 1$$

SC NMS A and B both correct 3/3One of A or B correct 1/3

(A1, A1)

3

$$(ii) \int \frac{7x - 3}{(x+1)(3x-2)} dx =$$

$$p \ln(x+1) + q \ln(3x-2)$$

Condone missing brackets

M1

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$$= 2 \ln(x+1) + \frac{1}{3} \ln(3x-2) (+c)$$

 F on A and B ; constant not required

A1F

2

$$(b) \frac{6x^2 + x + 2}{2x^2 - x + 1} = \frac{6x^2 - 3x + 3 + 4x - 1}{2x^2 - x + 1}$$

M1

$$= 3 + \frac{4x - 1}{2x^2 - x + 1}$$

$$P = 3$$

B1

$$Q = 4 \text{ and } R = -1$$

A1

Alternative:

$$\begin{array}{r}
 3 \\
 2x^2 - x + 1 \overline{) 6x^2 + x + 2} \\
 \underline{6x^2 - 3x + 3} \\
 4x - 1
 \end{array}$$

Complete division, with $ax + b$ remainder

(M1)

$$P = 3 \text{ stated}$$

(B1)

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

or

$$\begin{aligned}
 6x^2 + x + 2 &= P(2x^2 - x + 1) + Qx + R \\
 &= 2Px^2 + (Q - P)x + P + R
 \end{aligned}$$

Multiply across and equate coefficients or use numerical values of x

(M1)

$$P = 3$$

$$P = 3 \text{ stated}$$

(B1)

$$Q - P = 1$$

$$P + R = 2$$

$$Q = 4 \text{ and } R = -1$$

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

3

[8]

Q3.

(a) $\int \frac{dx}{\sqrt{x+1}} = \int -\frac{1}{5} dx$



Correct separation; or $\frac{dt}{dx} = -5(x+1)^{\frac{1}{2}}$
 Condone missing integral signs

B1

$$2\sqrt{x+1} = -\frac{1}{5}t \quad (+C)$$

Correct integrals; condone $\frac{\sqrt{x+1}}{\frac{1}{2}}$

B1B1

$$x = 80 \quad t = 0 \quad C = 2\sqrt{81}$$

Use (0, 80) to find a constant C

M1

$$= 18$$

F on integrals if in form $\sqrt{x+1} = qt + c$

A1F

$$x = \left(9 - \frac{1}{10}t\right)^2 - 1$$

OE; CSO; x = correct expression in t

A1

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(b) $t = 60 \quad x = f(60)$

Evaluate $f(60)$, ie $x = \dots$ (C not required)

M1

$$= 8$$

CSO

A1

2

(c) (i) $\frac{dA}{dt} = kA(9 - A)$

$\frac{dA}{dt}$ = product involving A; k required
 Condone terms in t

M1A1

2

(ii) $4.5 = \frac{9}{1 + 4e^{-0.09t}}$

Condone one slip in denominator

M1

$$e^{-0.09t} = \frac{1}{4}$$

A1

$$-0.09t = \ln\left(\frac{1}{4}\right)$$

Take ln correctly

m1

$$t = \frac{\ln\left(\frac{1}{4}\right)}{-0.09}$$

$$= 15.4 \text{ (hours)}$$

CAO; condone more than 3sf if correct
15.40327068 Allow 15h 24m

A1

4

[14]

Q4.

- (a) (i) $p(2) = 8 + 2 - 10$
Must find p(2) NOT long division

M1

$$\Rightarrow p(2) = 0 \Rightarrow (x - 2) \text{ is factor}$$

Shown = 0 plus a statement

A1

2

- (ii) Attempt at long division (generous)
Obtaining a quotient $x^2 + cx + d$
or equating coefficients (full method)

M1

$$p(x) = (x - 2)(x^2 + 2x + 5)$$

a = 2, b = 5 by inspection B1, B1

A1

2

(b) (i) $\frac{dy}{dx} = 3x^2 + 1$

One term correct

M1

All correct – no + c etc

A1

When $x = 2 \quad \frac{dy}{dx} = 3 \times 4 + 1$

Sub $x = 2$ into their $\frac{dy}{dx}$

m1

Therefore gradient at Q is 13

CSO

A1

4

(ii) $y = 13(x - 2)$

*Tangent (NOT normal) attempted
ft their gradient answer from (b)(i)*

M1

CSO; correct in any form

A1

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2

(iii) $\int \dots dx = \frac{x^4}{4} + \frac{x^2}{2} - 10x (+c)$

one term correct

M1

second term correct

A1

all correct (condone no +c)

A1

3

(iv) $[4 + 2 - 20] - [0] = -14$

F(2) attempted and possibly F(0)

Must have earned M1 in (b)(iii)

M1

Area of shaded region = 14

CSO; separate statement following correct evaluation of limits

A1

2

[15]

Q5.

(a) (i) $\frac{dy}{dx} = 3x^{\frac{1}{2}}$
 $kx^{\frac{1}{2}}$ with or without + c

M1

= 6 {when $x = 4$ }
Must be 6 and seen in (a)(i)
6 + c is A0

A1cao

2

(ii) y-coordinate of A = $2 \times 4^{\frac{3}{2}} (= 16)$

Substitute $x = 4$ in $y = 2x^{\frac{3}{2}}$

M1

$6 \times m' = -1$

$m_1 \times m_2 = -1$ OE used with c's value of

$\frac{dy}{dx}$ when $x = 4$. PI

M1

$y - 16 = m(x - 4)$

dep on 1st M1 in (a)(ii)

m must be numerical

m1



$$y - 16 = -\frac{1}{6}(x - 4)$$

ACF

A1

4

(b) (i)

$$\int 8x^{\frac{1}{2}} dx = \frac{8}{\frac{3}{2}} x^{\frac{1}{2}+1} \{+c\}$$

Index raised by 1

M1

$$= \frac{16}{3} x^{\frac{3}{2}} \{+c\}$$

Condone missing '+ c'
Coefficient must be simplified

A1

2

(ii)

$$\int 2x^{\frac{3}{2}} dx = \frac{2}{\frac{5}{2}} x^{\frac{5}{2}} \{+c\} \quad \{= \frac{4}{5} x^{\frac{5}{2}} \{+c\}\}$$

Can award for unsimplified form

B1

$$\int_0^4 8x^{\frac{1}{2}} dx - \int_0^4 2x^{\frac{3}{2}} dx$$

Ignore limits here

M1

$$= \frac{16}{3} (4)^{\frac{3}{2}} - 0 - \left[\frac{4}{5} (4)^{\frac{5}{2}} - 0 \right]$$

$F(4) - F(0)$ used in either; $\{F(0) = 0 \text{ PI}\}$
Cand. must be using $F(x)$ as a result of
his/her integration in (b)(i) or in the (b)(ii)
B1 line above

M1

$$= \frac{256}{15}$$

Accept any value from 17.04 to 17.1
inclusive in place of 256/15

A1

4

(c) Translation

*Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]*

B1

$$\begin{bmatrix} -3 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided
linked to 'translation/move/shift'
(B0B0 if >1 transformation)*

B1

2

[14]

Q6.

(a) (i) $\int \frac{dx}{e^{\frac{1}{2}x}} = \int -kt \, dt$

Separate; condone missing integral signs

B1

$$-2e^{-\frac{1}{2}x} = -k \frac{t^2}{2} (+C)$$

B1B1

3

(ii) $-2e^{\frac{1}{2}x} = -k \frac{t^2}{2} - 2e^{-3}$

Use (6, 0) to find constant

M1

$$\ln \left(e^{\frac{1}{2}x} \right) = \ln \left(k \frac{t^2}{4} + e^{-3} \right)$$

*Take logarithms correctly; condone one
side negative. Must have a constant.*

M1



$$-\frac{1}{2}x = \ln\left(k\frac{t^2}{4} + e^{-3}\right)$$

$$x = -2\ln\left(\frac{kt^2}{4} + e^{-3}\right)$$

Answer given; CSO

A1

3

(b) (i) $t = 10 \quad x = -2\ln\left(\frac{0.004 \times 10^2}{4} + e^{-3}\right)$

M1

$$= 3.8 \Rightarrow 3800$$

CAO

A1

2

(ii) $x = 0 \quad \frac{0.004 \times t^2}{4} + e^{-3} = 1$

M1

$$t = 30.8$$

CAO

Treat 0.04 or 0.0004 as misread (– 1)

A1

2

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[10]

Q7.

(a) $(n =) -4$

Accept x^{-4}

B1

1

(b) $\left(1 + \frac{3}{x^2}\right)^2 = 1 + \frac{6}{x^2} + \frac{9}{x^4}$

Apply ISW after B2 stage

(B1 if correct but unsimplified seen)

B2,1,0

2

(c) $\int \left(1 + \frac{3}{x^2}\right)^2 dx = x - 6x^{-1} - 3x^{-3} + c$

At least one power of x correctly obtained in the integration of an expansion

M1

*A2 terms correct **and** “+ c ”*

(A1F two terms in x correct ft on expansion provided integrating x to a negative power)

A2, 1, 0

3

(d) $\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 dx = \left[x - \frac{6}{x} - \frac{3}{x^3}\right]_1^3$
 $= \left(3 - \frac{6}{3} - \frac{3}{27}\right) - \left(1 - 6 - 3\right)$

*Dealing correctly with limits; $F(3) - F(1)$
 (must have attempted integration to get F)*

M1

$= 8\frac{8}{9}$

CSO;

$\frac{80}{9}, \frac{240}{27};$

*OE provided value is **exact**, eg $\frac{80}{9}, \frac{240}{27}$;
 ISW dec value after exact value seen
 NMS scores 0/2*

A1

2

[8]

Q8.

(a) $\int x dx = \int 150 \cos 2t \, dt$

Correct separation; condone missing $\int$ signs; must see dx, dt

B1

$\frac{1}{2}x^2 = 75 \sin 2t \quad (+C)$

Correct integrals



Accept $\frac{1}{2} \times 150$

B1B1

$$\left(20, \frac{\pi}{4}\right) \quad \frac{1}{2} \times 20^2 = 75 \sin\left(2 \times \frac{\pi}{4}\right) + C$$

C present. Use $\left(20, \frac{\pi}{4}\right)$ to find C

M1

$$C = 125$$

F on $x^2 = k \sin 2t$

A1F

$$x^2 = 150 \sin 2t + 250$$

Correct integrals and evaluation of C

A1

6

(b) (i) $t = 13 \quad x^2 = 150 \sin 26 + 250 (= 364.38)$

Evaluate $x^2 = f(13)$; $x^2 = k \sin 2t + c$
with numerical k and t

M1

$$x = 19.1 \text{ (cm)}$$

AWRT

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A1

2

$$(ii) \quad \left. \begin{array}{l} x = 11 \quad \sin 2t = -\frac{129}{150} \quad (= -0.86) \\ \text{or} \quad 2t = -1.035 \dots, 4.176 \dots \end{array} \right\}$$

M1

$$t = 2.1 \text{ (seconds)}$$

AWRT

A1

2

[10]

Q9.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
*Shown to = 0 **plus statement***

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$p(x) = (x + 1)(x + 2)(x - 3)$
*May earn M1, A1 For correct second factor
 then A1 For $(x + 1)(x + 2)(x - 3)$*

A1

3

(b) (i) $A(-2, 0)$
Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$

One term correct

M1

Another term correct

A1

(may have $+ c$ or not)
All correct unsimplified

A1

$$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q10.

(a) (i) $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$



A power decreased by 1

M1

A1 For each correct term

A1 A1

3

(ii) At $P(4, 0)$, $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts $\frac{dy}{dx}$ when $x = 4$

M1

$$= 1 - 3 = -2$$

AG

A1

2

(iii) Gradient of normal = $\frac{1}{2}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 0 = m[x - 4]$

m numerical; can be awarded even if

$m = -2$

M1

$$y - 0 = \frac{1}{2}(x - 4) \Rightarrow 2y = x - 4$$

ACF of the equation

A1

3

(iv) At Q , $x = 0$, $2y = 0 - 4$

PI

M1

$$y_e = -2$$

Area of triangle $OPQ = 0.5 \times 4 \times |y_e|$

Ft on a linear equation for normal provided y_e is negative and prev A1 is lost

A1F

$$= 4$$

Ft on c's negative y_e

B1F

3

$$(v) \quad 2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$$

Puts c's $\frac{dy}{dx} = 0$ and a 1st step in attempt to solve.

M1

$$2 = \frac{3}{2}x \Rightarrow x = \frac{4}{3}$$

Valid method to $ax = b$

m1

Condone 1.3 or better

A1

3

$$(b) \quad (i) \quad \int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$$

$$= \frac{8}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} \{+c\}$$

One power correct

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M1

Condone absence of "+ c"

A1,A1

3

$$(ii) \quad \text{Area under curve} = 4 \frac{4^{\frac{3}{2}}}{1.5} - \frac{4^{\frac{5}{2}}}{2.5} - \{0\}$$

$$F(4) - \{F(0)\}$$

Total area = $F(4)$ + area triangle OPQ

M1

m1



$$\text{Total area} = \frac{128}{15} + 4 = \frac{188}{15} = 12.5(3\dots)$$

Accept 3 sf if clear

A1

3

[20]

Q11.

(a) (i) $y = 2x^2 - 8x + 3$

$$\left(\frac{dy}{dx}\right) = 4x - 8$$

B1

1

(ii) $\int_4^6 \frac{x-2}{2x^2-8x+3} dx$
 $= \frac{1}{4} [\ln |2x^2 - 8x + 3|]_4^6$

M1 for $k \ln (2x^2 - 8x + 3)$; allow $k \ln u$

M1A1

$$= \frac{1}{4} [\ln 27 - \ln 3]$$

Correct substitution into

$k \ln (2x^2 - 8x + 3)$ or 3, 27 into $k \ln u$

m1

$$= \frac{1}{4} \ln 9$$

$$= \frac{1}{2} \ln 3$$

A1

4

(b) $\int x\sqrt{3x-1} dx$

$$u = 3x - 1 \quad du = 3 dx$$

OE

B1



$$\int = \left(\frac{1}{9}\right) \int \left(u^{\frac{3}{2}} + u^{\frac{1}{2}}\right) (du)$$

[2 terms in with rational indices

M1

$$= \left(\frac{1}{9}\right) \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} (+c) \right]$$

Must be 2 terms with correct indices

$$\left(\text{only fit for } x = \frac{u-1}{3} \right)$$

A1F

$$= \frac{2}{45} (3x-1)^{\frac{5}{2}} + \frac{2}{27} (3x-1)^{\frac{3}{2}} + c$$

CSO OE

A1

4

[9]

Q12.

$$\int y \, dy = \int 3 \cos 3x \, dx$$

attempt to separate and integrate

$py^2 = q \sin 3x$ seen $\Rightarrow$ implies separation

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$$\frac{1}{2} y^2 = \sin 3x (+C)$$

integrals – accept $\frac{1}{3} \times 3 \sin 3x$

A1A1

$$\left(\frac{\pi}{2}, 2\right) \frac{1}{2} \times 4 = \sin \frac{3\pi}{2} + C$$

use $\left(\frac{\pi}{2}, 2\right)$ to find constant

M1

$$C = 3$$

$$y^2 = 2 \sin 3x + 6$$

CSO (in any correct form)

A1

[5]

Q13.

(a) $\text{Grad } AC = \frac{15}{3} = 5$
OE

B1

Equation of AC: $y = m(x + 2)$

$$\text{or } (y - 15) = m(x - 1)$$

Or use of $y = mx + c$ with $(-2, 0)$ or

(1, 15) correctly substituted for x and y

M1

$$y = 5x + 10$$

$$\text{OE eg } y - 15 = 5(x - 1), y = 5(x + 2)$$

A1

3

(b) (i) $\left[16x - \frac{x^5}{5} \right]$

Raise one power by 1

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One term correct

A1

All correct

A1

$$\left(16 - \frac{1}{5} \right) - \left(-32 + \frac{32}{5} \right)$$

F(1) - F(-2) attempted

m1

$$= 41\frac{2}{5} \quad (\text{or } 41.4, \frac{207}{5} \text{ etc})$$

CSO; withhold if + c added

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 15$ or $22\frac{1}{2}$ or 22.5

Or $\int_{-2}^1 (5x + 10) dx = 22.5$

B1

Shaded area = "their (b)(i) answer" – correct triangle

Condone "difference" if $\Delta > \int$

M1

$\Rightarrow$ shaded area = $18\frac{9}{10}$

CSO; OE (18.9 etc)

A1

3

[11]

Q14.

(a) $\left(1 + \frac{4}{x^2}\right)^3 = \left[1^3\right] + 3(1^2)\left(\frac{4}{x^2}\right) + 3(1)\left(\frac{4}{x^2}\right)^2 + \left[\left(\frac{4}{x^2}\right)^3\right]$

Any valid method as far as term(s) in $1/x^2$ and term(s) in $1/x^4$

M1

$= [1] + \frac{12}{x^2} + \frac{48}{x^4} + \left[\frac{64}{x^6}\right]$

$p = 12$ Accept $\frac{12}{x^2}$ even within a series

A1

$q = 48$ Accept $\frac{48}{x^4}$ even within a series

A1

3

(b) (i) $\int \left(1 + \frac{4}{x^2}\right)^3 dx = \int \left(1 + \frac{p}{x^2} + \frac{q}{x^4} + \frac{64}{x^6}\right) dx$

*Integral of an 'expansion', at least 3 terms
PI by the next line*

M1

$$= x - px^{-1} - \frac{q}{3}x^{-3} - \frac{64}{5}x^{-5} (+c)$$

$$= x - 12x^{-1} - 16x^{-3} - \frac{64}{5}x^{-5} (+c)$$

At least two powers correctly obtained

m1

*Ft on c's non-zero integer values for p and q (A1F for two terms correct; can be unsimplified)
Condone missing c but check that signs have been simplified at some stage before the award of both A marks.*

A2F,1

4

(ii) $\left(2 - \frac{p}{2} - \frac{q}{3(8)} - \frac{64}{5(32)}\right) - \left(1 - p - \frac{q}{3} - \frac{64}{5}\right)$

F(2) – F(1), where F(x) is cand's answer or the correct answer to (b)(i).

M1

= 33.4

CSO

EXAM PAPERS PRACTICE

A1

2

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[9]

Q15.

(a) $\sin(x + 2x) = \sin x \cos 2x + \cos x \sin 2x$

M1

$= \sin x(1 - 2 \sin^2 x) + \cos x(2 \sin x \cos x)$

double angles; ACF ISW condone missing x

B1B1

$= \sin x(1 - 2 \sin^2 x) + 2 \sin x(1 - \sin^2 x)$

all in sin x, correct expression

A1



$$= 3 \sin x - 2 \sin^3 x - 2 \sin^3 x$$

$$= 3 \sin x - 4 \sin^3 x$$

CSO AG

A1

5

(b) $\sin^3 x = a \sin x + b \sin 3x$

attempt to solve for $\sin^3 x$ where $a \neq 0$ and $b \neq 0$

M1

$$\int \sin^3 x \, dx = -a \cos x - \frac{b}{3} \cos 3x$$

either integral correct

F on a, b

A1F

$$\int \sin^3 x \, dx = \frac{1}{4} \left(-3 \cos x + \frac{1}{3} \cos 3x \right) (+C)$$

CAO

alternative method by parts (see end of mark scheme)

A1

ALTERNATIVE METHOD 1

$$\int \sin^3 x \, dx = \int \sin^2 x \sin x \, dx$$

identify parts and attempt to integrate

M1

$$= -\sin^2 x \cos x - \int -2 \cos x \sin x \cos x \, dx$$

$$= -\sin^2 x \cos x - \frac{2}{3} \cos^3 x \quad (+C)$$

A2

ALTERNATIVE METHOD 2

$$\int \sin^3 x \, dx = \int \sin^2 x \, d(-\cos x)$$

condone sign error

M1



$$= \int -(1 - \cos^2 x) d(\cos x)$$

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

ALTERNATIVE METHOD 3

$$\int \sin x \sin^2 x dx$$

$$\int \sin x (1 - \cos^2 x) dx$$

this form and attempt to integrate

M1

$$= -\cos x + \frac{1}{3} \cos^3 x (+C)$$

A2

3

[8]

Q16.(a) (i) $B(0, 5)$

B1

Area AOB $\frac{1}{2} \times 1 \times 5$ *Condone slip in number or a minus sign*

M1

$$= 2 \frac{1}{2}$$

A1

3

$$(ii) \quad \frac{3x^6}{6} + \frac{2x^2}{2} + 5x \quad \text{or} \quad \frac{x^6}{2} + x^2 + 5x \quad (\text{may have } + c \text{ or not})$$

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

3

(iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$$[0] - \left[\frac{1}{2} + 1 - 5 \right]$$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

Area of shaded region = $3\frac{1}{2} - 2\frac{1}{2} = 1$
FT their integral and triangle (very generous)

B1 ✓

4

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

(ii) $y = \text{"their gradient"} (x + 1)$
*Must be finding **tangent** – not normal
 any form e.g. $y = 17x + 17$*

B1 ✓

1

[14]

Q17.

(a) (i) $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

*Power $p \rightarrow p - 1$
(A1 if $1 + ax^n$ with $a = -8$ or $n = -3$)*

A2,1,0

3

(ii) $1 - 8x^{-3} = 0$

Puts c's $\frac{dy}{dx} = 0$

M1

$x^3 = 8$

Using $x^{-k} = \frac{1}{x^k}$ to reach $x^a = b$, $a > 0$ or correct use of logs.

m1

$x = 2$

A1

When $x = 2$, $y = 4$

A1ft

4

(iii) At (1, 6), $\frac{dy}{dx} = 1 - 8 = -7$

Attempt to find $y'(1)$

M1

Gradient of normal = $\frac{1}{7}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 6 = m[x - 1]$

m numerical

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y-6}{x-1} = \frac{1}{7}; 7y = x + 4 \right\}$$

OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded

Alft

4

(b) (i) $\int x \left(1 + \frac{4}{x^2} \right) dx =$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+c\}$$

One of three terms correct.

M1

*One of three terms correct.
For A2 need all three terms as printed or better
(A1 if 2 of 3 terms correct)*

A2,1,0

3

(ii) {Area = } $\int_1^4 x + 1 + \frac{4}{x^2} dx =$
 $\left[\frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left(\frac{1}{2} + 1 - 4 \right)$

*Dealing correctly with limits; F(4)–F(1)
(must have integrated)*

M1

$$= 13.5$$

A1

2

[16]

Q18.

(a) (i) $\int (4 - e^{2x}) dx$

$$= 4x - \frac{1}{2}e^{2x} (+c)$$

$$4x$$

B1

$$-\frac{1}{2}e^{2x}$$

B1

2

$$(ii) \int_0^{\ln 2} \left[4x - \frac{1}{2}e^{2x} \right] dx$$

$$= \left[4 \ln 2 - \frac{1}{2}e^{2 \ln 2} \right] - \left[(0) - \frac{1}{2}(e^0) \right]$$

Substitute both $\ln 2$ and 0 correctly into an integrated expression

M1

$$= 4 \ln 2 - 2 + \frac{1}{2}$$

Convincing

$$= 4 \ln 2 - \frac{3}{2}$$

AG

A1

2

(b) (i) $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At B, $y = 0$

$$4 - e^{2x} = 0$$

Or reverse argument

M1

$$e^{2x} = 4$$

$$x = \ln 2$$

AG



A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

B1

$$x = \ln 2, \text{ Gradient} = -2e^{2\ln 2} = -8$$

$$x = \ln 2 \text{ into } ke^{2x}$$

M1

$$\text{Gradient normal} = \frac{1}{8} = \frac{1}{2e^{2\ln 2}}$$

OE

A1

$$\text{Equation } y = -\frac{1}{8}x - \frac{1}{8}\ln 2$$

OE

A1

4

(d) When $x = 0$

$$y = -\frac{1}{8}\ln 2$$

Attempt to integrate their line and substitute

$x = 0, \ln 2$

M1

$$\text{Area } \Delta = \frac{1}{16}(\ln 2)^2 \quad \text{condone - ve sign} = 0.03$$

$$\frac{1}{2} (\text{their } y) \times \ln 2$$

A1 ✓

$$\text{Total area } 4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.30$$

AWRT

CSO

A1

3

[14]

Q19.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$

Candidate's $\cos 2x$ or $\sin^2 x$

M1

$$= 3\sin x - 1 + 2\sin^2 x$$

AG

A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$

Soluble quadratic form

M1

$$(2\sin x - 1)(\sin x + 2) = 0$$

Attempt to solve (allow one error in formula, allow sign errors)

M1

$$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$$

M1

*$\sin^{-1}$ and two solutions ($0^\circ < x < 360^\circ$)
A0 if radians*

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$

Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$$x = 16.3^\circ, 163.7^\circ$$

Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)

M1A1

2

[9]

Q20.

(a) (i) $\int \frac{dy}{y} = \int \sin t \, dt$

Attempt to separate and integrate

M1

$\ln y = -\cos t + c$

A1 for $\ln y$; A1 for $-\cos t$; condone missing c

A1,A1

$y = ae^{-\cos t}$

A present; or $y = e^{-\cos t + C}$

A1

4

(ii) $y = 50, t = \pi: 50 = Ae^{-\cos \pi} = Ae$

Substitute $y = 50, t = \pi$ to find constant

Can have $50 = e^{1+C}$ if substituted in above

$e^C = \frac{50}{e}$

M1

A1

$y = 50e^{-1}e^{-\cos t}$

AG (convincingly obtained)

A1

3

Alternative:

Must have a constant in answer to (a)(i)



$$y = Ae^{-\cos t} \text{ or } y = e^{-\cos t + c} \text{ or } \ln y = -\cos t + c$$

$$50 = Ae^{-\cos \pi} \quad 50 = e^{-\cos \pi + c} \quad \ln 50 = -\cos \pi + c$$

(M1)

$$50 = Ae^{-1} \quad 50 = e^{1+c} \quad \ln y = -\cos t + \ln 50 - 1$$

(A1)

$$y = 50e^{-1-\cos t} \quad y = e^{-\cos t} \quad \frac{50}{e} \ln\left(\frac{y}{50}\right) = -1 - \cos t$$

(A1)

Alternative:Substitute $y = 50$, $t = \pi$ into $\ln y = -\cos t + c$ M1 $\ln y = -\cos t + \ln 50 - 1$ A1

$$\ln \frac{y}{50} = -1 - \cos t \quad (AG) \quad A1$$

(b) (i) $t = 6: y = 50e^{-1-\cos 6} = 7.0417... \approx 7\text{cm}$

Degrees 6.8 SC1

7 or 7.0 for A1

M1A1

2

(ii) $t = \pi \Rightarrow (\sin t = 0 \Rightarrow) \frac{dy}{dt} = 0$

Condone x for t

EXAM PAPERS PRACTICE

B1

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$$\frac{d^2 y}{dt^2} = y \cos t + \frac{dy}{dt} \sin t$$

For attempt at product rule including $\frac{dy}{dt}$ term; must have $\frac{d^2 y}{dt^2} =$

M1

$$t = \pi$$

A1



$$\frac{d^2 y}{dt^2} = y \cos \pi + \frac{dy}{dt} \sin \pi$$

$$= -50 \Rightarrow \max$$

Accept = -y, with explanation that y is never negative

A1

4

Alternative:

$$y = 50e^{-(1+\cos t)} = \frac{50}{e} e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e} e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2 y}{dt^2} = \frac{50}{e} e^{-\cos t} \times \cos t + \frac{50}{e} e^{-\cos t} \times \sin^2 t$$

Attempt at product rule

(M1)

Correct

(A1)

$$\text{Substitute } t = \pi \rightarrow -50 \Rightarrow \max$$

(A1)

EXAM PAPERS PRACTICE

[13]

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Q21.

(a) (i) $f(1) = 1 + 4 - 5$

must find f(1) NOT long division

M1

$$\Rightarrow f(1) = 0 \Rightarrow (x - 1) \text{ is factor}$$

shown = 0 plus a statement

A1

2

(ii) Attempt at $x^2 + x + 5$

*long division leading to $x^2 \pm x \dots$ or
equating coefficients*

M1

$$f(x) = (x - 1)(x^2 + x + 5)$$

$p = 1, q = 5$ by inspection scores B1, B1

A1

2

(iii) $(x =)1$ is real root

B1

Consider $b^2 - 4ac$ for their $x^2 + x + 5$
not the cubic!

M1

$$b^2 - 4ac = 1^2 - 4 \times 5 = -19 < 0$$

Hence no real roots (or only real root is 1)
CSO; all values correct plus a statement

A1

3

(b) (i) $\int \dots dx = \frac{x^2}{4} + 2x^2 - 5x(+c)$

one term correct unsimplified

M1

second term correct unsimplified

A1

all correct unsimplified

A1

3

(ii) $[4 + 8 - 10] - \left[\frac{1}{4} + 2 - 5 \right]$

*correct use of limits 1 and 2;
 $F(2) - F(1)$ attempted*

M1

$$= 4 \frac{3}{4}$$

A1

Area of $\Delta = \frac{1}{2} \times 11 = 5\frac{1}{2}$
correct unsimplified

B1

$\Rightarrow$ shaded area $= 5\frac{1}{2} - 4\frac{3}{4}$
*combined integral of $7x - 6 - x^3$ scores
 M1 for limits correctly used then*

$= \frac{3}{4}$

A3 correct answer with all working correct

A1

4

[14]

Q22.

(a) (i) x^2

B1

1

(ii) $x^{\frac{1}{2}} = \sqrt{x}$

Accept either form

B1

1

(iii) x^3

B1

1

(b) (i) $\int 3x^{\frac{1}{2}} dx = \frac{3}{\frac{3}{2}} x^{\frac{3}{2}} \{+c\}$

Index raised by 1

M1

Simplification not yet required

A1

$= 2x^{\frac{3}{2}} + c$



Need simplification and the + c OE

A1

3

$$(ii) \int_1^9 3x^{\frac{1}{2}} dx = (2 \times 9^{\frac{3}{2}}) - (2 \times 1^{\frac{3}{2}})$$

$F(9) - F(1)$, where $F(x)$ is candidate's answer to (b)(i) [or clear recovery]

M1

$$= 52$$

Ft on (b)(i) answer of form $kx^{1.5}$ i.e. $26k$

A1ft

2

[8]

Q23.

$$(a) \int x e^{5x} dx$$

$$u = x \quad dv = e^{5x}$$

integrate one term, differentiate one term

M1

$$du = 1 \quad v = \frac{1}{5} e^{5x}$$

A1

$$\int = \frac{1}{5} x e^{5x} - \int \frac{1}{5} e^{5x} dx$$

A1

$$= \frac{1}{5} x e^{5x} - \frac{1}{25} e^{5x} (+c)$$

A1

4

$$(b) \quad (i) \quad u = x^{\frac{1}{2}}$$

$$du = \frac{1}{2} x^{-\frac{1}{2}} dx$$

M1

$$\int = \int \frac{1}{1+u} \times 2du$$

correct with no errors; AG

A1

2

(ii) $\int_1^9 dx = \int_1^3 \frac{2}{1+u} du$

correct limits used in correct expression, ignoring k

m1

$$= [2 \ln(1+u)]_1^3$$

for k ln(1+u)

M1

$$= 2 \ln 4 - 2 \ln 2$$

(= ln 4)

ISW OE

A1

3

EXAM PAPERS PRACTICE

[9]

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Q24.

(a) $\tan x(+c)$

B1

1

(b) $f(x) = \frac{\cos x}{\sin x}$

$$f'(x) = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$$

quotient rule $\frac{\pm \sin^2 x \pm \cos^2 x}{\sin^2 x}$

M1

A1

$$= \frac{-1}{\sin^2 x}$$

use of $\sin^2 x + \cos^2 x = 1$

A1

$$= -\operatorname{cosec}^2 x$$

AG CSO

A1

4

Special cases

$$f(x) = \frac{\cot x}{1}$$

$$f'(x) = \frac{1 \times -\operatorname{cosec}^2 x - \cot x \times 0}{1^2}$$

M1

$$= -\operatorname{cosec}^2 x$$

A1

(max 2/4)

Or

$$f(x) = \frac{1}{\tan x}$$

$$f'(x) = \frac{\tan x \times 0 - 1 \times \sec^2 x}{\tan^2 x}$$

M1A1

$$= \frac{-\sec^2 x}{\tan^2 x}$$

$$= \frac{-1}{\sin^2 x} = -\operatorname{cosec}^2 x$$

A1

(max 3/4)

(c) LHS = $\tan^2 x + \cot^2 x + 2 \tan x \cot x$

expanding

M1

$$= \tan^2 x + 1 + \cot^2 x + 1$$

correct use of trig identities

M1

$$= \sec^2 x + \operatorname{cosec}^2 x$$

CSO

A1

3

= RHS

(d) $\int (\tan x + \cot x)^2 dx = \int \sec^2 x + \operatorname{cosec}^2 x dx$
use of identity

M1

$= [\tan x - \cot x]_{0.5}^1$
 $\pm \tan x \pm \cot x$ OE

M1
A1

$= 0.9153 - -1.2842$

$= 2.2$

AWRT

A1

4

[12]

Q25.

(a) $\int \frac{1}{\sqrt{1+2y}} dy = \int \frac{1}{x^2} dx$
attempt to separate and integrate

M1

$\int \frac{1}{\sqrt{1+2y}} dy = k\sqrt{1+2y}$

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m1

$\sqrt{1+2y} = -\frac{1}{x} (+c)$

OE A1 for $\sqrt{1+2y}$ depends on both Ms

A1

A1 for $-\frac{1}{x}$ depends on first M1 only

A1

$x = 1, y = 4 \Rightarrow c = 4$



+ c must be seen on previous line

m1

ft on k and $\pm \frac{1}{x}$ only

A1F

6

(b) $1+2y=\left(4-\frac{1}{x}\right)^2$

need $k\sqrt{1+2y} = 'x \text{ expression with } + c'$
and attempt to square both sides

m1

$$2y=15+\frac{1}{x^2}-\frac{8}{x}$$

terms on RHS in any order
AG – convinced CSO

A1

2

[8]

Q26.

(a) $y_D = 3 + 1 = 4$ or $y_C = 12 - 8 = 4$

Attempt at either y coordinate

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M1

Area $ABCD = 3 \times 4 = 12$

A1

2

(b) (i) $x^3 - \frac{x^4}{4} \quad (+ C)$

Increase one power by 1

M1

One term correct unsimplified

A1

All correct unsimplified (condone no +C)

A1

3



- (ii) Sub limits -1 and 2 into their (b)(i) ans
May use both $-1, 0$ and $0, 2$ instead

M1

$$[8 - 4] - \left[-1 - \frac{1}{4} \right] = 5 \frac{1}{4}$$

A1

Shaded area = "their" (rectangle – integral)
Alt method: difference of two integrals

M1

$$= 12 - 5 \frac{1}{4} = 6 \frac{3}{4}$$

CSO. Attempted M2, A2

A1

4

(c) (i) $\frac{dy}{dx} = 6x - 3x^2$
One term correct

M1

All correct (no +C etc)

A1

2

- (ii) When $x = 1, y = 2$ when $x = 1,$
May be implied by correct tgt equation

B1

$$\frac{dy}{dx} = 3 \text{ as 'their' grad of tgt}$$

Ft their derivative when $x = 1$

M1ft

Tangent is $y - 2 = 3(x - 1)$
Any correct form $y = 3x - 1$ etc

A1

3

- (iii) Decreasing when $\frac{dy}{dx} = 6x - 3x^2 < 0$



Watch no fudging here!!
May work backwards in proof.

M1

$$3(2x - x^2) < 0 \Rightarrow x^2 - 2x > 0$$

AG (be convinced no step incorrect)

A1

2

- (d) Two critical points 0 and 2

Marked on diagram or in solution

M1

$$x > 2, \quad x < 0 \quad \text{ONLY}$$

or M1 A0 for $0 < x < 2$ or $0 > x > 2$

SC B1 for $x > 2$ (or $x < 0$)

A1

2

[18]

Q27.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

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A1

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2

(b) (i) When $x = 0$, $\frac{dy}{dx} = -3$

Ft provided answer < 0 .

B1F ✓

Eqn of tangent at O is $y = -3x$

OE Ft on $y'(0)$

B1F ✓

2

(ii) At $(9,0)$ $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find $y'(9)$

M1

Eqn tangent at A is $y - 0 = y'(9)[x - 9]$

OE

m1

$$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$$

CSO. AG

A1

3

(iii) Eliminating $y \Rightarrow -6x = 3x - 27$

OE method to one variable

(eg $2y = -y - 27$)

M1

$$9x = 27 \Rightarrow x = 3$$

[A1F for each coordinate; only ft on $y = kx$ tangent
in (b)(i) for $k < 0$]

A1F

When $x = 3$, $y = -9$. $P(3, -9)$

A1F

3

(c) $\int \left(x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$

One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx =$
PI

B1



$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

Correct use of limits following integration

M1

$$= -24.3$$

$$\text{Area of triangle } OPA = \frac{1}{2} \times 9 \times |y_P|$$

M1

$$\text{Sh. Area} = \frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx \right|$$

OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

Q28.

(a) $\cos 2x = 2 \cos^2 x - 1$

B1B1

2

(b) $\cos^2 x = \frac{1}{2} (\cos 2x + 1)$

Attempt to express $\cos^2 x$ in terms of $\cos 2x$

M1

$$\frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x + 1 \, dx = \left[\frac{1}{4} \sin 2x + \frac{x}{2} \right]_0^{\frac{\pi}{2}}$$

A1

A1

$$= \frac{\pi}{4}$$

Use limits. Ft on integer a .

Q29.

(a) (i) $\frac{dy}{dx} = 3x^2 - 20x + 28$

One term correct

M1

Another term correct

A1

All correct (no + c etc)

A1

3

(ii) Their $\frac{dy}{dx} = 0$ for stationary point
Or realising condition for stationary point

M1

$(x - 2)(3x - 14) = 0$
Attempt to solve using formula/ factorise

m1

$\Rightarrow x = 2$

A1

or $x = \frac{14}{3}$

*Award M1, A1 for verification that $x = 2 \Rightarrow \frac{dy}{dx} = 0$
 then may earn m1 later*

A1

4

(b) (i) $\frac{x^4}{4} - \frac{10x^3}{3} + 14x^2 (+ c)$

One term correct unsimplified

M1

Another term correct unsimplified

A1

All correct unsimplified (condone missing + c)

A1

3

(ii) $\left[\frac{81}{4} - 90 + 126 \right] \quad (-0)$

Attempt to sub limit 3 into their (b)(i)

M1

$= 56\frac{1}{4}$

AG Integration, limit subtraction all correct

A1

2

(iii) Area of triangle = $31\frac{1}{2}$

Correct unsimplified $\frac{1}{2} \times 21 \times 3$

B1

Shaded Area = $56\frac{1}{4} - \text{triangle area}$

M1

$= 24\frac{3}{4}$

Or equivalent such as $\frac{99}{4}$

B1

3

[15]

Q30.

(a) (i) When $x = 4$, $\frac{dy}{dx} = 3(2) + \frac{16}{16} - 7 = 0$

AG Be convinced

B1

1

(ii) $\frac{16}{x^2} = 16x^{-2}$
Accept $k = -2$

B1

1

(iii) $\frac{d^2y}{dx^2} = 3 \times \frac{1}{2}x^{-\frac{1}{2}} + 16 \times (-2)x^{-3} - 0$
A power decreased by 1

M1

$$\frac{d^2y}{dx^2} = \frac{3}{2}x^{-\frac{1}{2}}; -32x^{-3}$$

candidate's negative integer k [-1 for >2 term(s)]

A1;
 A1 ✓

3

(iv) When $x = 4$, $\frac{d^2y}{dx^2} = \frac{3}{4} - \frac{32}{64} = \frac{1}{4}$
Attempt to find $y''(4)$ reaching as far as two simplified terms

M1

Minimum since $y''(4) > 0$

candidate's sign of $y''(4)$

E1 ✓

2

[Alternative: Finds the sign of $y'(x)$ either side of the point where $x = 4$, need evidence rather than just a statement: (M1)

Correct ft conclusion with valid reason E1 ✓]

[In both, condone absent statement $y'(4) = 0$]

(b) (i) At $P(1,8)$, $\frac{dy}{dx} = 3(1)^{\frac{1}{2}} + \frac{16}{1^2} - 7 = 12$
AG Be convinced

B1

1

(ii) Gradient of normal = $-\frac{1}{12}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 8 = m[x - 1]$

Can be awarded even if $m = 12$

M1

$$y - 8 = -\frac{1}{12}(x - 1) \Rightarrow 12y - 96 = -x + 1 \Rightarrow 12y + x = 97$$

Any correct form of the equation

A1

3

$$\int 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7dx = \dots\dots\dots = 3\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 16\frac{x^{-1}}{-1} - 7x + c$$

(c) (i)

One power correct.

M1

A1 if 2 of 3 terms correct candidate's negative integer k

Condone absence of "+ c"

A2,1,0 ✓

3

(ii) $y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + c$ (*)

y = candidate's answer to (c)(i) with tidied coefficients and with '+c'. ('y =' PI by next line)

B1 ✓

When $x = 1, y = 8 \Rightarrow 8 = 2 - 16 - 7 + c$

Substitute. (1,8) in attempt to find constant of integration

M1

$$y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + 29$$

Accept $c = 29$ after (), including $y =$, stated*

A1

3

[17]

Q31.

(a) $9x^2 - 6x + 5 = 3(3x - 1)(x - 1) + A(x - 1) + B(3x - 1)$

Or $3 + \frac{6x + 2}{(3x - 1)(x - 1)}$

B1

$x = 1 \quad x = \frac{1}{3}$

Substitute $x = 1$ or $x = \frac{1}{3}$

M1

$B = 4 \quad A = -6$

Or equivalent method (equating coefficients, simultaneous equations)

A1A1

4

(b) $\int = \int 3 - \frac{6}{3x - 1} + \frac{4}{x - 1} dx$

Attempt to use partial fractions

M1

$= 3x \dots$

B1

$-2 \ln(3x - 1) + 4 \ln(x - 1) (+c)$

$p \ln(3x - 1) + q \ln(x - 1)$

Condone missing brackets

M1

Follow through on A and B; brackets needed.

A1F

4

[8]

Q32.

$$\int \frac{dy}{y^2} = \int 6x \, dx$$

Attempt to separate
Either dx or dy in right place

M1

$$-\frac{1}{y} = 3x^2 (+C)$$

$$-\frac{1}{y}; 3x^2$$

A1A1

$$x = 2 \quad y = 1 \quad C = -13$$

Use (2, 1) to find a constant.

M1

CAO

A1

$$y = \frac{1}{13 - 3x^2}$$

CAO OE

A1

6

[6]

Q33.

- (a) (i) $x - 1$ is a factor

May be earned retrospectively

B1

$x + 2$ is a factor

From part (iii)

B1

2

- (ii) Attempt at third factor

Multiplying/ dividing/factor theorem

M1

$$f(x) = (x - 1)(x + 2)(x - 4)$$

$$(x+4) \Rightarrow M1, A0$$

A1

2

- (b) (i) At A , $y = 8$
Or $(0, 8)$

B1

1

- (ii) At B , $x = 4$
Or $(4, 0)$ NO ft of wrong factor

B1

1

- (c) (i) $\frac{x^4}{4} - x^3 - 3x^2 + 8x \quad (+c)$

Increase one power by 1

M1

One term correct (unsimplified)

A1

Two other terms correct (unsimplified)

A1

All correct (unsimplified)
(condone missing + c)

A1

4

- (ii) Realisation that limits are -2 and 1
Condone wrong way round

B1

$$\text{Area} = \left[\frac{1}{4} - 1 - 3 + 8 \right] - [4 + 8 - 12 - 6]$$

Attempt to sub their limits into their (c)(i)

M1

$$= 20 \frac{1}{4}$$

CSO. Must use $F(1) - F(-2)$ correctly

A1

3

Q34.

(a) $\sqrt{x} = x^{\frac{1}{2}}$

Accept $k = 0.5$

B1

1

(b) $\sqrt{x}(x-1) = x^{\frac{1}{2}}x - x^{\frac{1}{2}} = x^{\frac{3}{2}} - x^{\frac{1}{2}}$

M1

Accept $p = 1.5, q = 0.5$

A1

2

(c) $\int \sqrt{x}(x-1) dx = \frac{x^{2.5}}{2.5} - \frac{x^{1.5}}{1.5} (+c)$
Increases a power of x by 1

M1

ft non-integer p

A1ft

ft non-integer q

A1ft

3

(d) $\int_1^2 dx = \left(\frac{2^{2.5}}{2.5} - \frac{2^{1.5}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$
Limits; $F(2) - F(1)$

M1

$\dots = \left(\frac{4\sqrt{2}}{2.5} - \frac{2\sqrt{2}}{1.5} \right) - \left(\frac{1}{2.5} - \frac{1}{1.5} \right)$

Fractional powers to surds

m1

$\left(\frac{24\sqrt{2}}{15} - \frac{20\sqrt{2}}{15} \right) - \left(-\frac{4}{15} \right) = \text{pr. ans}$

CSO AG (be convinced)

A1

3

[9]

Q35.

(a) (i) $p(3) = 27 - 45 + 21 - 3$
Finding $p(3)$

M1

$p(3) = 0 \Rightarrow x - 3$ is a factor
Shown = 0 plus a statement
Or $(x - 3)(x^2 - 2x + 1)$

A1

2

(ii) B is point (3,0)
Must have coordinates

B1

1

(b) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$
One term correct

M1

All correct with NO + c etc

A1

2

(ii) $3x^2 - 10x + 7 = 0$

Putting their $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$
Attempt to use quad formula or factorise

m1

$\Rightarrow$ at M , $x = \frac{7}{3}$
CSO factors correct etc

A1

3

(c) $\frac{d^2 y}{dx^2} = 6x - 10$

ft their $\frac{dy}{dx}$

B1ft

sub $x = 1, \Rightarrow \frac{d^2 y}{dx^2} = -4$

ft their $\frac{d^2 y}{dx^2}$

B1ft

2

(d) (i) $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x$ (+c)
Increase one power by 1

One term correct

M1

A1

Two other terms correct

A1

All correct (condone missing + c)

A1

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4

- (ii) Realisation that limits are 0 and 1
Condone wrong way round

B1

$$\left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3 \right] - 0$$

Attempt to sub their limits into their (d)(i)

M1

$$= -\frac{11}{12}$$

CSO. *Must use $F(1) - F(0)$ correctly*



A1

$$\text{Area} = \frac{11}{12}$$

CSO. *Convincing argument*

E1

4

[18]



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